

Ramana Nadapana

Senior Lead Data Engineer | Databricks Technical Consultant

Address: Houston, Texas, USA 77058

Phone: +1 346-433-7728 | **Email:** nssvramana@gmail.com

Professional Summary

Data Engineer with 10+ years of experience in designing, implementing, and optimizing large-scale data pipelines using Azure Analytics Platform and Databricks.

Expertise in designing and implementing Azure cloud-based solutions with strong proficiency in Azure Fabric, One Lake, and cloud data architecture.

Proven track record in designing and deploying data lake/delta lake architectures, ETL pipelines using Azure Databricks and Azure Data Factory, and modern data warehouses like Azure Synapse Analytics.

Specialization in Spark-based data processing, Delta Lake for transactional storage, and advanced ETL/ELT workloads.

Strong programming skills in Python and Scala, with expertise in Spark development to build scalable and efficient data processing solutions.

Developed and optimized SQL queries with business logic to efficiently fetch and process data, reducing query execution time by 20%.

Skilled in ensuring data quality, governance, and compliance using Databricks Delta, Unity Catalog, and scalable monitoring frameworks.

Experienced in ETL solution implementations using Azure Data Flow, Azure Databricks, AWS Glue and AWS Lambda.

Implemented data pipelines using the medallion architecture, ensuring streamlined data flow and enhancing data accessibility for downstream analytics.

Expertise in data warehouse solutions such as Snowflake and Azure Synapse Analytics, optimizing data storage and query performance to support real-time business insights.

SME support for troubleshooting, analysis and resolutions on production issues on existing workloads.

Strong experience in CI/CD, DevOps and Airflow for automating data pipeline deployments.

Passionate about mentoring junior engineers and enabling data-driven transformations within enterprises.

Certifications

- Azure Certified Data Engineer - Associate Application
- Google Certified Professional Data Engineer
- AWS Certified Data Engineer – Associate
- CDP Certified Administrator - Private Cloud Base

Experience

HSBC India Development Center, India

Senior Lead Data Engineer

08/2022 - 02/2025

- Designed and implemented scalable data pipelines and ETL processes using Azure Databricks, Azure Data Lake Gen2, Azure Data Factory to streamline data integration for business insights.

- Expertise in implementing Azure Fabric-based data architectures with One Lake integration to ensure scalable and cost-effective cloud data platforms.
- Built Delta Lake tables using SCD Type 2 for effective data storage, enabling accurate historical data tracking and improved data consistency.
- Developed Azure-based pipelines that automated data integration and transformation processes, improving data accuracy and reducing manual effort by 40%.
- Managed data governance frameworks with Unity Catalog, ensuring compliance, data security, and efficient access control in cloud environments.
- Exported business data to Postgres for downstream visualization and reporting, enhancing decision-making for stakeholders.
- Built ETL solutions using Airflow, achieving a 16% improvement in conversion rates through optimized data workflows.
- Collaborated with analytical teams to integrate finalized business data into BI tools, improving reporting capabilities.
- Led cross-functional teams to deliver high-impact solutions, reducing operational costs by 30%.
- Designed, developed, and deployed integration solutions, ensuring seamless data movement and transformation across platforms.
- Debug and troubleshoot issues during development and production phases.
- Experience in using CI/CD pipelines improving deployment efficiency and reducing errors.

Technologies/Tools: Azure Databricks, Spark, Pyspark, Unity Catalog, Azure data lake Gen2, Azure Data Factory, Azure Synapse Analytics, Airflow, Postgres and python and SQL.

L&T Technology Services

Senior Lead Data Engineer

09/2021 - 08/2022

- Designed and maintained scalable Azure ETL pipelines to deliver high impact Cloud Data Solutions.
- Optimized Apache Spark workloads in Databricks, improving large-scale data transformations by 25%.
- Orchestrated data pipeline execution using Apache Airflow DAGs, ensuring efficient scheduling, dependency management, and error handling, reducing data latency by 30%.
- Automated data quality checks using python and implemented the data governance protocols.
- Converted legacy SQL scripts to Spark transformations using Databricks notebooks, improving query performance and maintainability.
- Managed data warehouses in Azure Synapse analytics optimizing queries for improved performance.
- Experience in Azure Synapse analytics SQL pool management.
- Designing, Implementing, and maintain Continuous Integration and Continuous Deployment pipelines.
- Conducted unit testing, resolved bugs, and ensured timely delivery to the testing team, improving deployment success rates.
- Sprint wise implementation in and code review.
- Helping team for the functional or technical glitches if any.

Technologies/Tools: Azure Databricks, Spark, Pyspark, Unity Catalog, Azure data lake Gen2, Azure Data Factory and Azure Synapse analytics, Airflow, Postgres and python and SQL.

Atos India Limited

Senior Data Engineer

01/2021 - 08/2021

- Designed and maintained ADF pipelines to centralize data into Azure Data Lake Gen2.

- Optimized Spark jobs to enhance large-scale data processing performance, reducing data pipeline execution time by 30%.
- Orchestrated data pipeline execution using Apache Airflow, automating data quality checks to improve data accuracy and reduce manual interventions.
- Managed end insightful business data in modern warehouses to optimizing improved performance for Reporting BI tools.
- Understanding the requirements from the functional team and guiding the team in developing the technical specifications.
- Involved in all the business discussions along with the client.
- Supported existing production workloads, proactively identifying and resolving recurring issues through root cause analysis.

Technologies/Tools: Azure Databricks, Azure Data Factory, Azure Data Lake Gen2, Spark, Pyspark, Airflow, Python, SQL

Infosys Limited

Bigdata Developer

08/2016 - 12/2020

- Built ETL pipelines using Hadoop MapReduce, Hive, and Spark for large-scale data processing.
- Supported the maintenance and optimization of spark and hive workload to improve job processing speed.
- Designed and optimized data models in Azure Synapse analytics.
- Experience in migration of data and existing workloads from on-prem HDFS to Azure blob storage.
- Integrated on-premises data sources with cloud platforms using Talend and Informatica.
- Experience in supporting and handling production job failures during data migration from on-prem to Azure cloud.
- Developed batch processing pipelines to improve business intelligence and analytics workflows.
- Provided BI solutions in Power BI and Tableau, supported business decision-making.

Technologies/Tools: Hadoop, Spark, Hive, HDFS, Azure blob storage, Azure data factory, HiveQL, SQL and python

Infosys Limited

Bigdata Platform Engineer

02/2015 - 08/2016

- Expertise in maintenance and optimization of spark and hive workload to improve job performance.
- Design, build and maintain the Cloudera Hadoop clusters for various environments.
- Handled the production issues for various Hadoop issues.
- Managed petabytes (PBs) of HDFS data in Cloudera clusters, optimizing storage and processing for large-scale data operations.
- Upgraded and patched Cloudera cluster stacks to the latest versions, ensuring optimal performance and security compliance.
- Integrated Cloudera clusters with different BI tools using JDBC/ODBC drivers.
- Coordinated with cross-functional stakeholders during major incident management, ensuring timely resolution and clear communication.

Technologies/Tools: Hadoop, HDFS, OOZIE, HBase, Spark, Hive, Knox, Ranger, shell scripting, SQL and python

Skills

- **Cloud Platforms:** Azure Databricks, Azure (Data Factory, Synapse Analytics, Data Lake), Azure (Azure Fabric, One Lake), AWS (Redshift, Glue, Lambda, S3, EMR)
- **Programming Languages:** Python, SQL, Yaml, Shell Scripting
- **Big Data Technologies:** Apache Spark, Hadoop, Kafka
- **Data Warehousing/Databases:** Hive, AWS Redshift, Azure Synapse Analytics, MySQL, PostgreSQL, Oracle
- **ETL Tools:** AWS Glue, ADF, Dataflow, Apache Airflow
- **DevOps & CI/CD:** Ansible, Jenkins
- **Version Control:** Git, GitHub, Bitbucket

Education

B.Tech. (Electrical and Electronics Engineering), 2014
Jawaharlal Nehru Technological University Kakinada (JNTUK) - India

Website, Portfolio and Profiles

<https://www.linkedin.com/in/ramana-nadapana-544b73157/>